APIRL 2026.

PROJECT REPORT

SALES DATA ANALYSIS & DASHBOARD – SUPERSTORE DATASET

Internship

Data Science and Analytics Internship

Organization

Future Interns

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Summary

- This project analyzes the Superstore sales dataset to extract key business insights using Python and Excel. Data was cleaned, processed, and visualized to understand sales, profit, category performance, and regional trends. The analysis revealed that Technology is the most profitable category, while the Western region performs best overall. An interactive Excel dashboard was created to present insights clearly and support data-driven decision-making.

Problem Statement

- Businesses generate large amounts of sales data, but it is often difficult to interpret and extract meaningful insights from raw data. Without proper analysis, it becomes challenging to identify profitable products, underperforming categories, regional performance differences, and sales trends over time.
  This project aims to analyze the Superstore sales dataset to uncover key patterns in sales and profit performance. The objective is to transform raw data into meaningful insights using data analysis and visualization techniques, enabling better business decision-making and performance improvement strategies.

Project Objectives

- To analyze the Superstore sales dataset to understand overall business performance.
- To identify sales and profit trends across different time periods.
- To evaluate the performance of product categories and sub-categories.
- To analyze region-wise sales and profit distribution.
- To identify top-performing and underperforming products and segments.
- To create clear visualizations and an interactive dashboard for better insight understanding.
- To support data-driven decision-making for improving business strategies and profitability

Data Description

The dataset used in this project is the Sample Superstore dataset, which contains:

- **9,994** records and **21** attributes
- Order details including Order Date and Ship Date
- Sales, Profit, Discount, and Quantity information
- Customer segmentation and geographical data (Region, State, City)
- Product classification (Category and Sub-Category)

Data Preparation

Data Cleaning & Processing

- Verified dataset integrity and checked for missing values (none found)
- Converted date fields into proper datetime format
- Ensured correct data types for numerical and categorical variables
- Prepared dataset for grouping and analysis using Python (Pandas)

Tools & Technologies Used

- Python: Data analysis and visualization
- Pandas: Data manipulation and aggregation
- Matplotlib: Graphical visualization
- Microsoft Excel: Dashboard creation and reporting

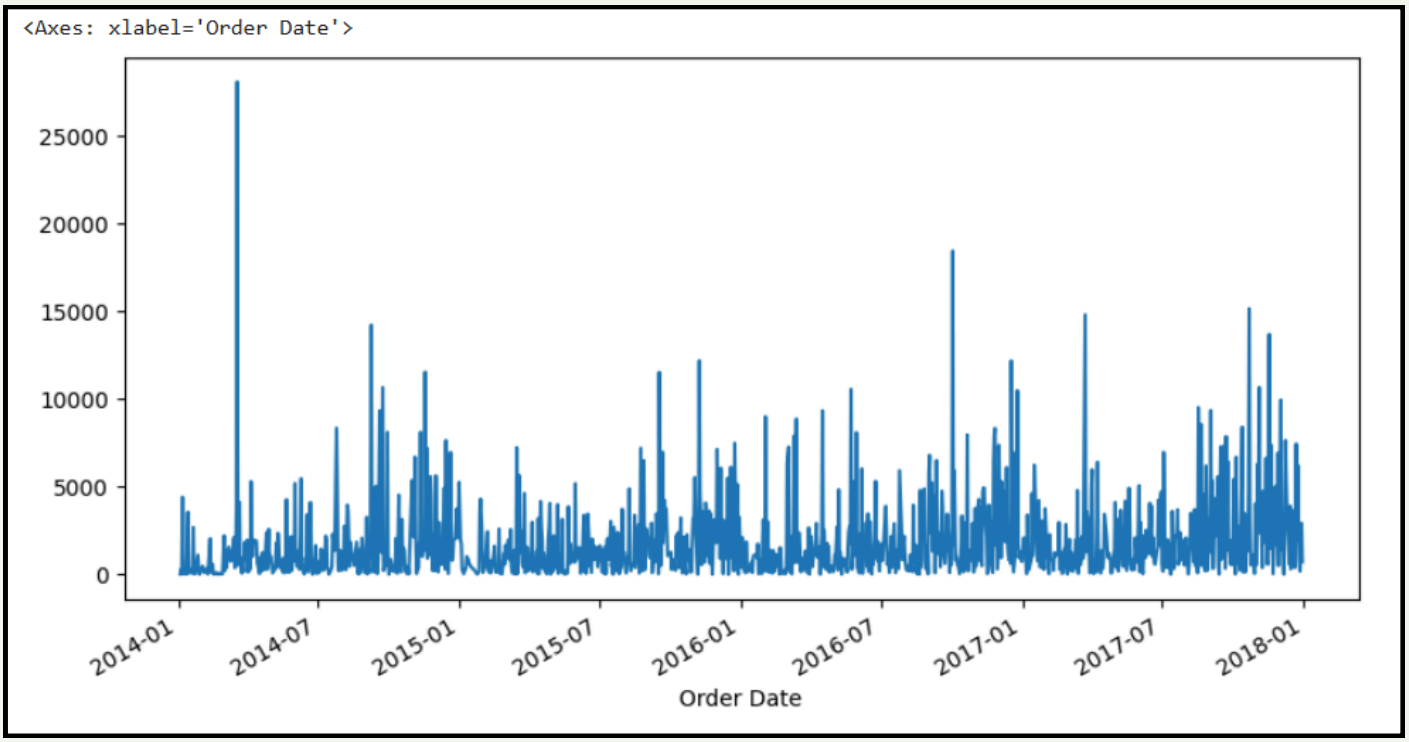
Work Flow

Start —→ Load Dataset —→ Clean Data —→ EDA —→ Analysis —→ Visualization —→ Dashboard —→ Insights —→ End

Exploratory Data Analysis (EDA)

1.Sales Trend Analysis

The time series shows a progressive upward trend in sales, indicating sustained business growth over the observed period. However, the growth is non-linear and volatile, with frequent short-term fluctuations.
Several sharp spikes are observed, likely driven by seasonal demand, promotional campaigns, or high-value transactions. The increase in spike frequency and magnitude during 2016–2018 suggests improved customer acquisition and business expansion. Additionally, the gradual rise in baseline sales indicates consistent underlying demand growth beyond peak events.



Exploratory Data Analysis (EDA)

2.Top Selling Products

The analysis of product performance was conducted based on both total sales and total profit to identify the most valuable products for the business.

- **Top Products by Sales**

The sales analysis shows that a limited number of products contribute significantly to overall revenue. Products such as **Canon imageCLASS 2200 Advanced Copier** and other office equipment dominate in terms of sales value, indicating strong customer demand.

Product Name	Sales
Canon imageCLASS 2200 Advanced Copier	61599.824
Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind	27453.384
Cisco TelePresence System EX90 Videoconferencing Unit	22638.480
HON 5400 Series Task Chairs for Big and Tall	21870.576
GBC DocuBind TL300 Electric Binding System	19823.479
GBC Ibimaster 500 Manual ProClick Binding System	19024.500
Hewlett Packard LaserJet 3310 Copier	18839.686
HP Designjet T520 Inkjet Large Format Printer - 24" Color	18374.895
GBC DocuBind P400 Electric Binding System	17965.068
High Speed Automatic Electric Letter Opener	17030.312

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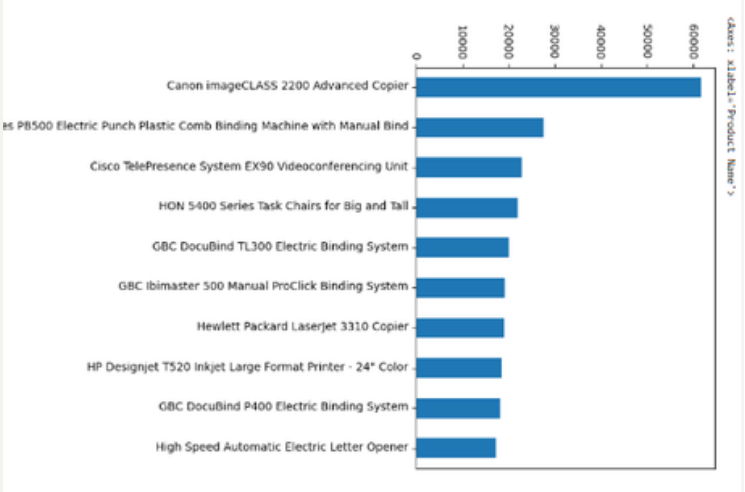

- **Top Products by Profit**

Profit-based analysis reveals that **Canon imageCLASS 2200 Advanced Copier** is also the **highest profit**-generating product. Several other products contribute moderate profits, but overall profitability is concentrated among a few key items.

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Product Name	Profit
Canon imageCLASS 2200 Advanced Copier	25199.9280
Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind	7753.0390
Hewlett Packard LaserJet 3310 Copier	6983.8836
Canon PC1060 Personal Laser Copier	4570.9347
HP Designjet T520 Inkjet Large Format Printer - 24" Color	4094.9766
Ativa V4110MDD Micro-Cut Shredder	3772.9461
3D Systems Cube Printer, 2nd Generation, Magenta	3717.9714
Plantronics Savi W720 Multi-Device Wireless Headset System	3696.2820
Ibico EPK-21 Electric Binding System	3345.2823
Zebra ZM400 Thermal Label Printer	3343.5360

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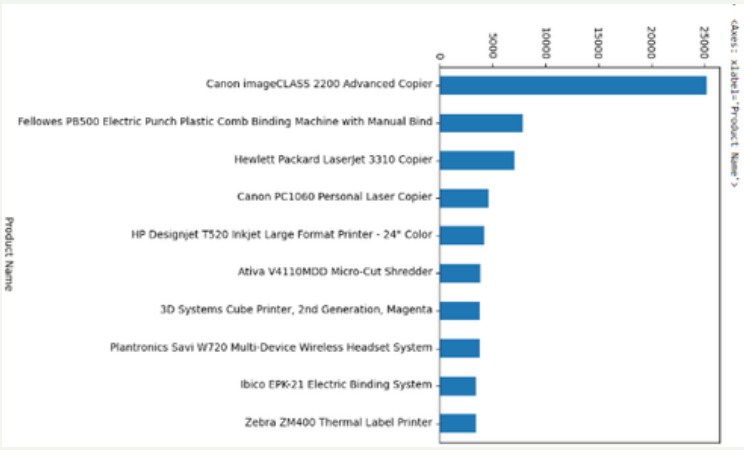

Exploratory Data Analysis (EDA)

3.Category Performance

The analysis shows that **Technology generates the highest revenue**, making it the most valuable category, while **Office Supplies records the highest order** volume, indicating strong demand. **Furniture** shows comparatively **lower profitability**, suggesting potential inefficiencies.

Category Count

Category	count
Office Supplies	6026
Furniture	2121
Technology	1847

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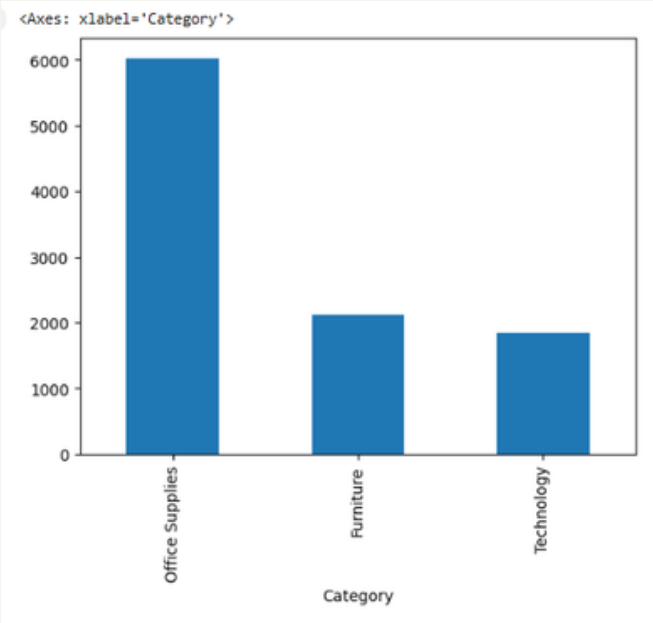

Category Sales

Category	Sales
Furniture	741999.7953
Office Supplies	719047.0320
Technology	836154.0330

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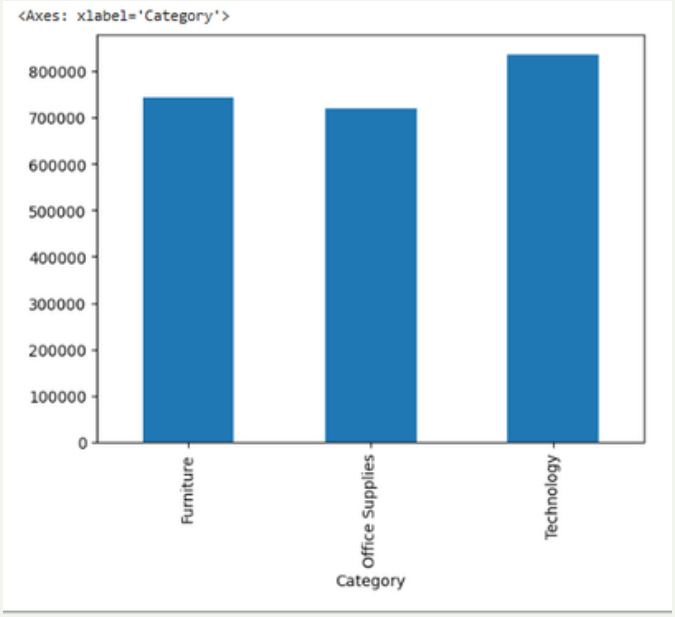

Exploratory Data Analysis (EDA)

4.Regional Sales Analysis

The analysis shows that the **West region** generates the **highest sales**, indicating strong market performance and **higher revenue** contribution. The **East region** follows as the **second-highest** contributor, while the **Central region** shows **moderate** sales levels. The **South region** records the **lowest sales**, suggesting comparatively weaker performance.

Regional Sales

Region	Sales
Central	501239.8908
East	678781.2400
South	391721.9050
West	725457.8245

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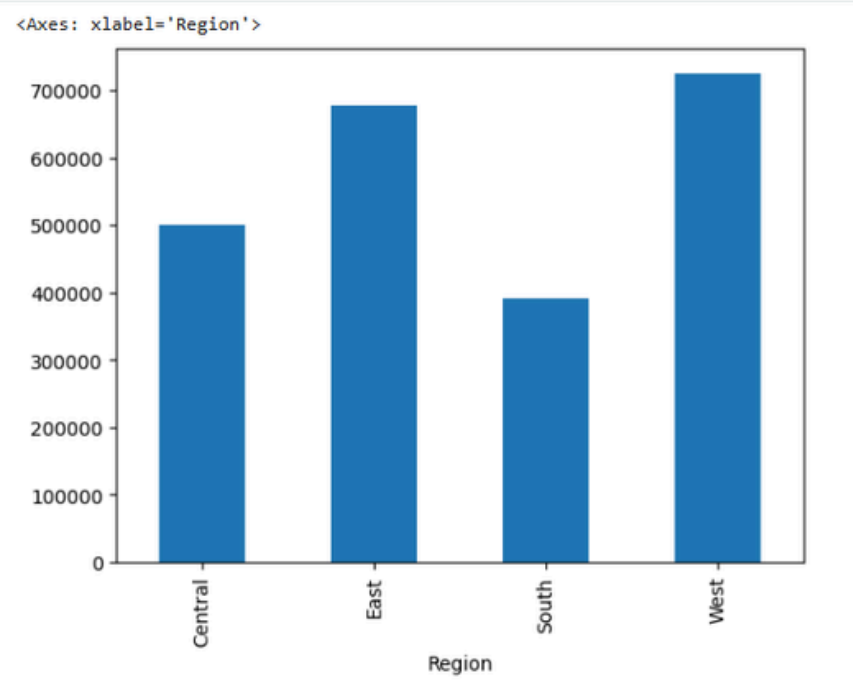
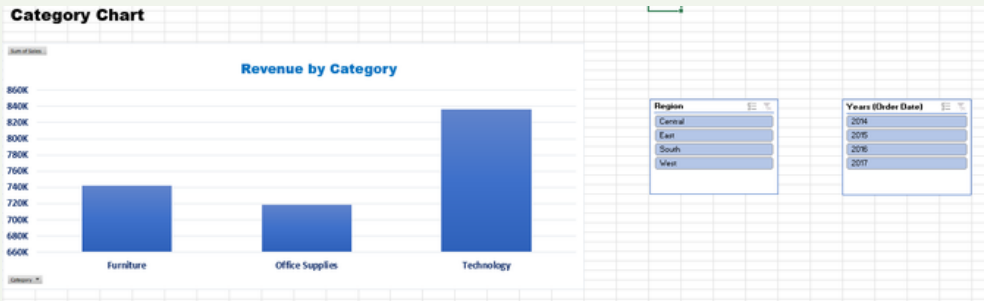
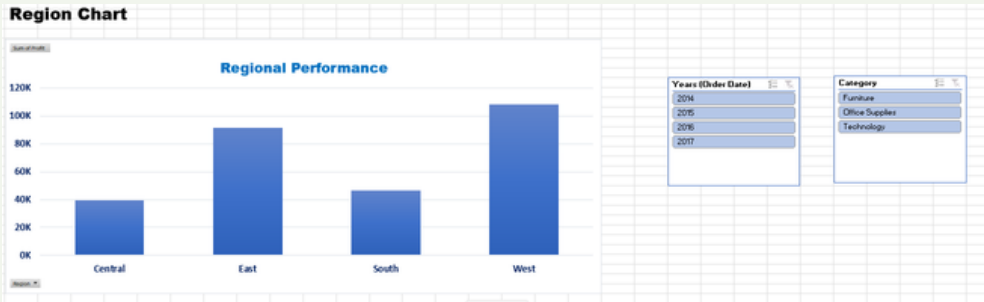


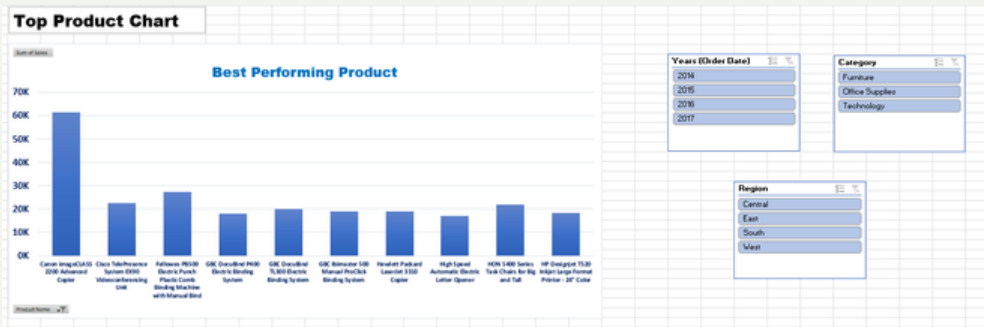
Dashboard

An interactive dashboard was developed using Microsoft Excel to provide a clear and concise visualization of key business metrics. The dashboard enables quick analysis of sales trends, product performance, category insights, and regional distribution.

- Sales Trend Analysis – Displays how sales change over time
- Top Products – Highlights highest revenue and profit-generating products
- Category Performance – Compares sales across categories
- Regional Analysis – Shows sales distribution across regions
- Interactive Filters (Slicers) – Allows dynamic data exploration







Key Insights

- Sales show an overall increasing trend with noticeable fluctuations
- A few products contribute significantly to both revenue and profit
- Technology is the highest-performing category in terms of sales
- West region leads in sales, while South shows lower performance
- Sales and profit are not always directly proportional, indicating margin variations

Business Recommendations

- Focus on high-profit products to maximize overall profitability
- Strengthen strategies in low-performing regions (South)
- Invest more in the Technology category to drive growth
- Optimize pricing and cost structure for low-margin products
- Leverage seasonal trends for better sales planning

Conclusion

This project demonstrates the effectiveness of data analysis in extracting valuable insights from raw sales data. By analyzing sales trends, product performance, and regional distribution, the study identifies key drivers of revenue and profitability, along with areas that require improvement. The insights and dashboard developed in this project support data-driven decision-making and provide a strong foundation for enhancing business performance and achieving sustainable growth.

Thankyou